# Amey Mishra

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## Education

**HARMONY SCHOOL OF ADVANCEMENT** | Houston, TX | Expected May 2027
**Valedictorian | GPA: 4.75 | SAT: 1540 | Intended Major: Mechanical/Aerospace Engineering**
**Advanced Coursework** — Multivariable Calculus, Linear Algebra, College Physics II, AP Calculus BC, AP Research, AP Cybersecurity, AP Environmental Science, AP English Literature.

## Technical Skills

**Mechanical / CAD** — Fusion 360, Onshape, Autodesk Revit, mechanism design, DFM, 3D printing, prototyping, actuator integration.
**Programming / Controls** — Python, Java, Arduino, JavaScript, OpenCV, sensor feedback, autonomous navigation, path planning, inverse kinematics.
**Research / Analysis** — Simulation, experimental design, data analysis, orbital mechanics, technical literature review, feasibility assessment.

## Engineering, Research and Leadership

**FOUNDER, CAPTAIN & MECHANICAL LEAD** | 06/2023 - Present
**FIRST Tech Challenge Team 25679 The Cosmobots | thecosmobots.com**

- Led robot architecture, mechanical design, prototyping, fabrication, testing, strategy, and team operations for an internationally competitive FTC program.
- Finished among the top teams worldwide out of 8,000+ teams; World Championship 4th-ranked alliance; Carolinas Premier Champion; **FIRST** Tech Challenge Hall of Fame.
- **FIRST** Leadership Award Finalist; reduced transfer time from ~0.75 s to ~0.30 s through iterative CAD, mechanism testing, actuator-placement studies, and serviceability reviews.
- Contributed to autonomous/control development through feedback loops, custom path-planning approaches, and mechanism libraries; coordinated design reviews, build timelines, scouting, and competition preparation.

**FOUNDER & CEO** | 06/2024 - Present
**Cosmobots Robotics | cosmobotsrobotics.com**

- Founded and scaled a STEM nonprofit supporting four FTC teams, three American Rocketry Challenge teams, technical mentorship, and community outreach.
- Secured $50K+ in sponsorships/grants/in-kind support; coordinated budgets, equipment, registrations, mentors, parent volunteers, travel, and program planning.
- Logged 4,500+ organizational service hours and 30+ demonstrations/workshops; reached 25,000+ people directly and students in seven countries; built partnerships with Unitree Robotics, Stokes **Education**, Harmony Public Schools, Meiborg, and others.

**LEAD RESEARCHER & SYSTEMS ENGINEER** | 2025 - Present
**Space Debris Research / NASA Space Apps - ORCA | orcadebriscleanup.space**

- Published research on orbital-debris growth, collision risk, operational impacts, and mitigation; led ORCA, a capture-and-recycling mission concept.
- Trained a debris-analysis neural network, applied orbital mechanics to capture-path planning, and CAD-designed/simulated collection and recycling mechanisms.
- Developed technical feasibility, operations, implementation, and business plans; NASA Space Apps Regional Winner and Global Semifinalist.

**INDEPENDENT ROBOTICS & AI RESEARCHER** | 2026 - Present
**Mentored by a Stanford PhD Student**

- Developed a neural-network-powered humanoid arm with custom CAD/3D-printed structure, motors, sensors, embedded control, and OpenCV perception.
- Implemented adaptive inverse kinematics and Java/Arduino controls; iterated actuator placement, joint geometry, sensing, and model behavior through physical testing and calibration.

**DESIGN & MANUFACTURING INTERN** | 07/2026 - Present
**Pike Robotics**

- Designed motion systems for an industrial tank-inspection robot, including pivoting camera mounts and umbilical-management mechanisms for tanks over 100 ft in diameter.
- Used Fusion 360 and Onshape to compare concepts around packaging, motion range, serviceability, reliability, and manufacturability.

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## Engineering and Professional Experience

**SUMMER RESEARCH INTERN** | Summer 2026
**Rice University - Electromagnetic Energy Harvesting**
- Investigated electromagnetic energy harvesting for low-power electronics; programmed microcontroller test systems and simulated prototype performance under varying conditions.
- Analyzed feasibility/tradeoffs, presented findings at the London International Conference, and submitted a research paper to the Society of North American Scholars.

**TEAM FOUNDER, CAPTAIN & PILOT** | 2023 - Present
**RECF Aerial Drone Competition - Teams 96610A / 96610B**
- Managed two teams, flight operations, FAA TRUST compliance, safety training, equipment, registration, practices, travel, and competition logistics.
- Built a reusable Python autonomous-flight library using sensor feedback; earned 3 Regional Championships, 3 Excellence/All-Around Awards, 3 Nationals qualifications, Flight Operations Award, and Judges Award.
- Mentored developing teams on flight control, programming, safety, and competition preparation.

**TEAM LEAD / FLIGHT SYSTEMS DEVELOPER** | 2025 - Present
**American Rocketry Challenge & Thrust-Vector-Control Development**
- Led rocket design, fabrication, testing, launches, budgeting, and team operations; designed/tested a custom thrust-vector-control system using CAD, 3D printing, electronics, and flight-data analysis.
- Built autonomous programs with ~97% successful execution in testing; pursuing supervised NAR Level 1 high-power rocketry certification.

**PRESIDENT & ENGINEERING EVENT COMPETITOR** | 2023 - Present
**Harmony School of Advancement Science Olympiad**
- Led a 15-member team and specialized in engineering/build events; earned 7 Regional Golds, 1 State Gold, and 2 State Bronzes in Tower, Air Trajectory, Flight, Helicopter, Electric Vehicle, and related events.

**STUDENT COUNCIL PRESIDENT** | 2023 - 2025
**Harmony School of Advancement**
- Represented ~800 students; led seven schoolwide events/service initiatives, coordinated officers and faculty sponsors, and organized Class of 2027 fundraising.

## Engineering Projects

**Autonomous Mapping UAV** — CAD-designed/3D-printed UAV with waypoint navigation, high-resolution camera, live video downlink, and iterative flight testing.
**RC Aerodynamics Test Aircraft** — Designed/3D-printed aircraft to test flaps, spoilers, stability, and aerodynamic behavior through repeated flight trials.
**Humanoid Robot** — Integrated Arduino motor control, OpenCV perception, API-based task planning, facial recognition, and kinematics-based movement.
**Autonomous Lawn Mower** — Integrated drive systems, sensing, electronics, and control logic around traction, packaging, and safety constraints.
**NASA Exoplanet Analysis** — Trained/refined a CNN on public NASA datasets to identify possible exoplanet candidates and improve classification consistency.
**3D Printing Services** — Self-employed custom fabrication work; translated client requirements into CAD, material, slicing, tolerance, and production decisions.

## Honors

**FIRST** — Leadership Award Finalist; Hall of Fame Team; Carolinas Premier Champion; World Championship 4th-ranked alliance.
**NASA / Aerospace** — NASA Space Apps Regional Winner & Global Semifinalist; Gene Kranz Scholarship recipient; Space Center U Thermodynamics event winner.
**STEM Competitions** — International Math Challenge Gold; RECF national Flight Operations & Judges Awards; Science Olympiad State Gold/2x Bronze; eCYBERMISSION Texas 2nd.

## Certifications & Additional Activities

**Certifications** — Autodesk Fusion 360; Autodesk Revit; CompTIA Network+; OpenEDG Python; Microsoft ITS Python; FAA TRUST; Adobe Photoshop/Illustrator; W3Schools JavaScript.
**Speedcubing** — Official WCA competitor; 9.38 s 3x3 and 2.07 s 2x2 personal bests; 2x school winner and 2x finalist.
**STEM Mentoring** — Mentored robotics, drone, Science Olympiad, and engineering students through design reviews, troubleshooting, programming support, and competition preparation.